

Section C: NAE-3-SAT Solution

Problem: NAE-3-SAT

Definition:

NAE-3-SAT asks whether a CNF formula with three literals per clause has an assignment where every clause contains at least one true literal and at least one false literal.

Task 1 Solution:

Example unsatisfiable NAE formula:

$(x_1 \vee x_2 \vee x_3) \wedge$
 $(\neg x_1 \vee \neg x_2 \vee x_3) \wedge$
 $(x_1 \vee \neg x_2 \vee \neg x_3) \wedge$
 $(\neg x_1 \vee x_2 \vee \neg x_3)$

Any assignment makes at least one clause all true or all false.

Task 2: NAE-3-SAT is in NP

Certificate:

Variable assignment.

Verification:

Evaluate every clause and check that each clause contains both true and false literals.

This can be done in polynomial time.

Task 3 and 4: Reduction from 3-SAT

Transform a 3-SAT formula by introducing additional variables and clauses so that satisfying assignments correspond exactly to NAE satisfying assignments.

The transformation increases formula size polynomially.

Correctness:

If the original 3-SAT formula is satisfiable, the transformed NAE-3-SAT formula is satisfiable.

If the transformed formula is satisfiable, the original formula must be satisfiable.

Therefore NAE-3-SAT is NP-complete.